Predicting Employee Attrition - IBM HR Analytics

Week 7 Graded Mini Project - Executive Summary | IITM Pravartak, Professional Certificate Programme in Generative AI and Machine Learning | Arpit Jain

Summary

A classification model was built to flag employees who are likely to leave, using the IBM HR Analytics dataset. Five models were tuned and compared. The final model is a **regularised Logistic Regression** with a tuned decision threshold. On unseen test data it ranks a random leaver above a random stayer 80% of the time (ROC-AUC 0.80). At its operating point, 58% of the people it flags really left, and it catches about half of all leavers. The clearest drivers of attrition are **overtime** (odds of leaving x3.4), **low pay and early career**, **stalled promotions**, **frequent travel** and **low job involvement**.

1. Dataset overview

Item	Value
Source	IBM HR Analytics Employee Attrition (Kaggle), fictional data created by IBM
Rows / columns	1,470 employees / 35 columns (26 numeric, 9 text)
Target	Attrition: Yes = 237 (16.1%), No = 1,233 (83.9%), imbalanced about 5:1
Missing values / duplicates	None / none
Dropped columns	EmployeeCount, Over18, StandardHours (constant), EmployeeNumber (ID)
Engineered features	AvgSatisfaction, TenureRatio, YearsPerCompany, PromotionStagnation, IncomeVsLevel
Split	Stratified 80/20 (1,176 train / 294 test), seed 42; test set used only for final evaluation

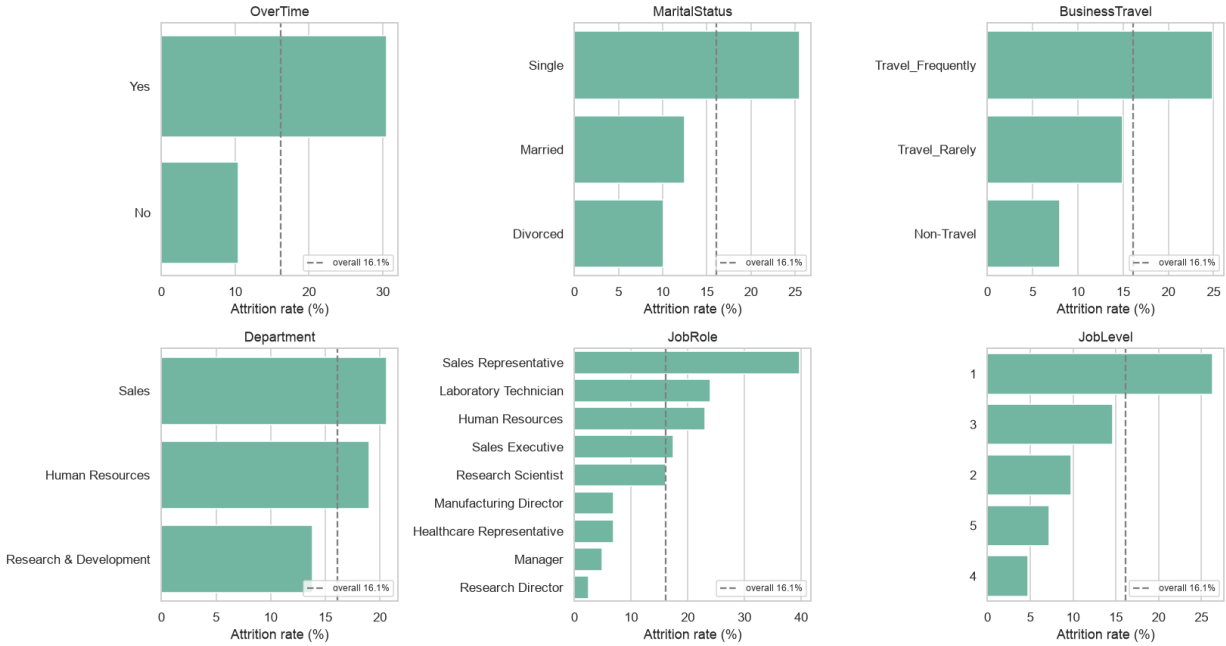


Figure 1 - Attrition rate by category (dashed line = 16.1% overall). Overtime, single status, frequent travel, Sales Representatives and job level 1 stand out.

2. Approach

- **EDA:** target balance, attrition rate per category, numeric distributions of leavers vs stayers, satisfaction ratings, a correlation matrix and an IQR outlier check.
- **Preprocessing** was done in one scikit-learn / imbalanced-learn pipeline: feature engineering, then one-hot encoding for nominal columns and standard scaling for numeric and ordinal columns. Because it is a single pipeline, every step is re-fitted inside each CV fold, so there is no data leakage.
- **Imbalance:** class weights for LR, Decision Tree, Random Forest and SVM; SMOTE (applied only to training folds) for Gradient Boosting.
- **Models:** Logistic Regression, Decision Tree, Random Forest, SVM (RBF) and Gradient Boosting. Each was tuned with GridSearchCV and 5-fold stratified CV, optimising F1 for the 'leave' class.
- **Threshold tuning:** each model's cut-off was chosen on out-of-fold training predictions, then applied once to the untouched test set.
- **Drivers:** logistic regression odds ratios, permutation importance on raw columns, and Random Forest importances as a cross-check.

3. Challenges faced

- **Class imbalance makes accuracy misleading.** Predicting 'nobody leaves' already scores 84% accuracy. Precision, recall, F1, ROC-AUC and PR-AUC for the leaver class were used instead.
- **Imbalance handling has a side effect.** Class weights raised recall at the 0.5 cut-off from 0.47 to 0.74, but F1 at 0.5 fell (0.58 to 0.50) and ROC-AUC did not change. The methods shift probabilities rather than improve ranking, so the threshold had to be re-tuned per model. Without that, the models could not be compared fairly.
- **Multicollinearity.** JobLevel and MonthlyIncome correlate at 0.95, and age, experience and tenure move together. This was handled with L2 regularisation and by reading drivers from permutation importance as well as from coefficients.
- **Outliers.** The IQR rule flags about 10% of rows, but these are genuine values (senior salaries, long tenure) or artefacts of discrete columns (PerformanceRating is only 3 or 4). They were kept, and their influence was limited through scaling and regularisation.
- **Small minority class.** With only 237 leavers, results move a few points between folds (CV F1 std about 0.04-0.06), so differences smaller than that were treated as ties.
- **Reloading the model.** Feature engineering lives in a separate module (attrition_features.py), so the saved pipeline can be loaded outside the notebook.

4. Model comparison

CV columns are from 5-fold cross-validation on the training set. Test columns are on the 294 held-out employees, at each model's own tuned threshold.

Model	CV F1 (@0.5)	CV ROC-AUC	Tuned threshold	Test Precision	Test Recall	Test F1	Test ROC-AUC	Test PR-AUC
Logistic Regression	**0.507**	**0.834**	**0.68**	**0.575**	**0.489**	**0.529**	**0.801**	**0.572**
SVM (RBF)	0.551	0.842	0.34	0.535	0.489	0.511	0.805	0.560
Gradient Boosting + SMOTE	0.551	0.825	0.40	0.458	0.468	0.463	0.785	0.526
Random Forest	0.491	0.789	0.52	0.426	0.489	0.455	0.755	0.376
Decision Tree	0.425	0.707	0.50	0.307	0.489	0.377	0.683	0.272

Why Logistic Regression? After threshold tuning, SVM and Logistic Regression are tied on the training data (out-of-fold F1 0.602 vs 0.593, a gap well within fold noise). On the test set, LR has the best F1 and PR-AUC, with ROC-AUC equal to SVM's. With performance tied, LR wins on explainability: HR can see why each person is flagged. Tree models were clearly weaker. A single decision tree underfits once it is constrained enough not to overfit.

Default vs tuned threshold for the final model: at 0.5 it flags 83 people and catches 30 of 47 leavers (precision 0.36). At the tuned 0.68 it flags 40 and catches 23 (precision 0.58, F1 0.46 to 0.53). The threshold is a business lever: lower it when retention actions are cheap, raise it when they are costly.

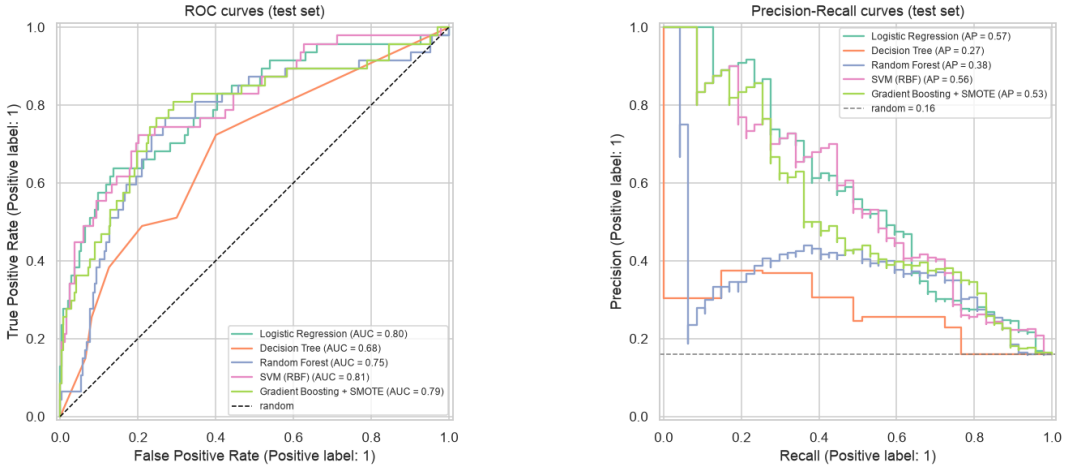


Figure 2 - ROC and precision-recall curves on the test set.

5. Key insights - what drives attrition

Driver	Evidence
Overtime	30.5% attrition vs 10.4%; odds ratio 3.4; top permutation importance
Low pay / early career	Median income 3,202 (leavers) vs 5,204; job level 1 at 26%; #2 permutation importance
Job hopping	Low years per previous employer: Random Forest's top feature, odds ratio 0.60 per s.d.
Stalled promotion	Years since last promotion (same tenure): odds ratio 1.65 per s.d.
Travel and commute	Frequent travellers 24.9% vs non-travel 8.0%; distance odds ratio 1.33
Low engagement	Job involvement rating 1: 33.7% vs rating 4: 9.0%; low job and environment satisfaction
Role	Sales Representative 39.8%, Lab Technician 23.9%; Research Director 2.5%
Marital status	Single 25.5% vs married 12.5% / divorced 10.1%
Protective	More years with the current manager (odds ratio 0.75)

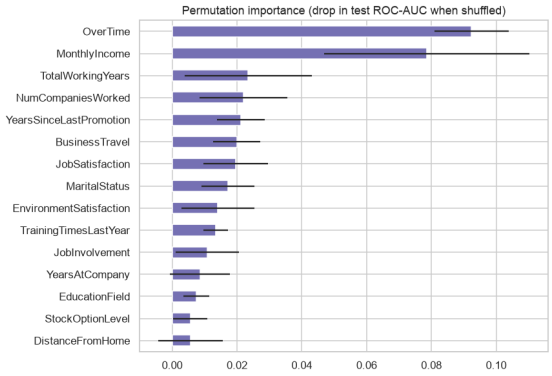


Figure 3 - Permutation importance of the final model (drop in test ROC-AUC when a column is shuffled).

6. Recommendations

- **Monitor and cap overtime.** It is the biggest single lever. Track overtime per team monthly, and rebalance workload or add headcount where it is chronic.
- **Review junior pay and promotion timelines.** Benchmark job-level-1 salaries, and flag anyone with no promotion in 3+ years for a career conversation.
- **Treat low engagement scores as an early warning.** Follow up when job involvement or satisfaction drops to 1-2.
- **Target retention programmes at Sales Representatives and Lab Technicians**: career paths, incentives and workload review.
- **Reduce travel and commute strain** by rotating travel-heavy assignments and offering hybrid work for long commutes.
- **Use the model as a quarterly risk-ranked watch-list** for HR business partners, not as an automatic decision. Measure whether interventions actually lower attrition.

7. Limitations and next steps

- The data is small (237 leavers) and fictional. The model should be re-validated on the company's own HR data.
- The drivers are associations, not proven causes; for example, single employees also tend to be younger and lower paid.
- Next steps: per-employee explanations (SHAP), a time-to-exit (survival) model, and probability calibration if the scores will be read as exact percentages.